

Elite Athletic AI Advisor

A youth-first coaching agent that turns athlete profiles into safe, coach-reviewable weekly development plans.

Ages 13-17

core youth MVP with parent and coach oversight

6

evaluation cases including two rejection examples

\$36k

ARR example per 250-athlete academy

Built in Databricks with Gold tables, tool-grounded agent logic, MLflow evaluation, and a two-LLM ROI comparison.

Business thesis

The winning product is not an unsupervised AI coach. It is a safer first-draft plan generator that keeps humans in the loop and compounds proprietary development data over time.

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Fragmented Guidance

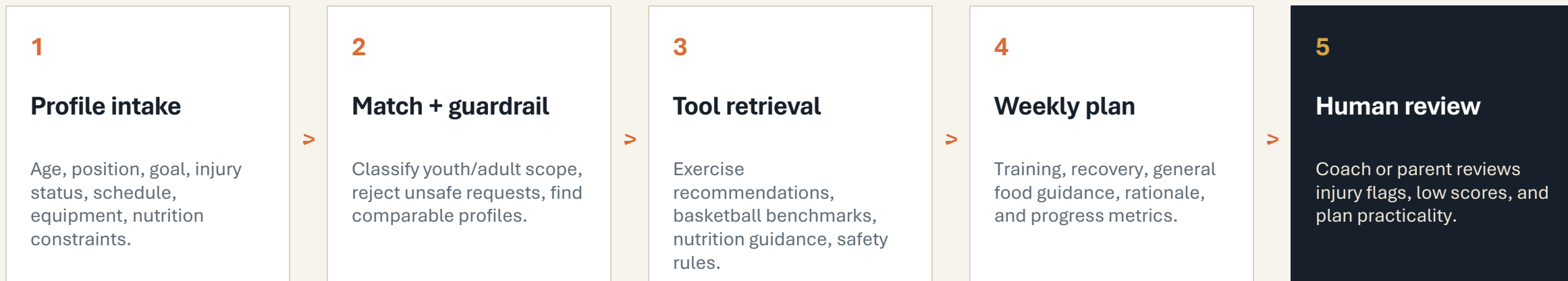
Youth players, parents, coaches, and academies all want better development decisions, but the advice loop is noisy and uneven.

Athlete	Parent	Coach	Academy
Current reality Generic drills, social media routines, and little feedback on what to do this week.	Current reality Wants ambition, but also worries about overtraining, unsafe diet advice, and injury risk.	Current reality Has more athletes than time for fully individualized weekly plans.	Current reality Needs a scalable product layer that improves retention without hiring linearly.
Product requirement Needs a plan that fits age, goal, equipment, and soreness.	Product requirement Needs age-aware guardrails and transparent reasoning.	Product requirement Needs a strong first draft to review and adjust.	Product requirement Needs measurable quality, repeatable delivery, and data capture.

The wedge is trust: safe, specific, explainable weekly guidance that a human can supervise.

Personalized Fitness Plan

The agent should reduce planning friction while leaving accountability with parents, coaches, and qualified clinicians for injury concerns.



Positioning line for the video: "Elite Athletic AI does not replace coaching judgment. It makes safe, individualized first drafts cheap enough to give every athlete one."

Personalization starts with a profile, that gets matched with professional athlete profiles.

This is the product layer to emphasize: users should enter their own details, then the agent matches them to similar athlete patterns before coaching.

Example raw user input

"I am 16, play shooting guard, want explosiveness and ball handling, have knee soreness sometimes, can train 4 days a week, and have dumbbells plus a driveway hoop."

Output: a structured profile that the tools can validate, score, match, and trace.

Profile fields

Age and age group
Sport and position
Skill level and goals
Injury or soreness notes
Days available
Equipment access
Nutrition goal
Oversight model

Matching score

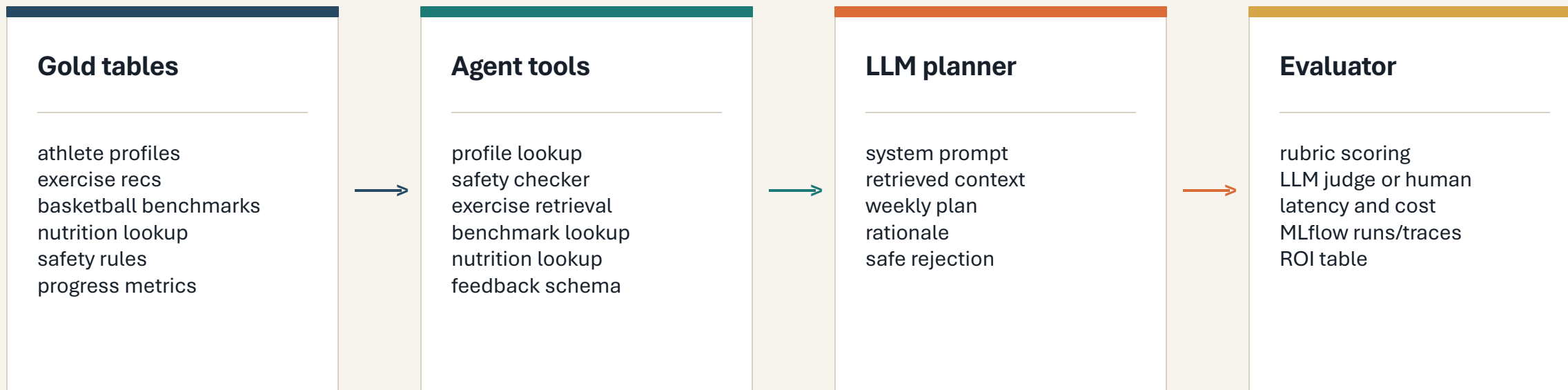


Top 3 matches become the rationale for training intensity, exercise selection, and safety notes.

Presentation point: the current POC proves the agent loop; the next implementation step should make this intake-to-match path explicit.

The agent is structured by guardrails and database retrieval, not just prompting.

The implementation uses Databricks tables and custom tools so the LLM is grounded in profile, training, nutrition, benchmark, and safety context.



This satisfies the agent requirement: LLM + multiple tools + safety/error handling + evaluation across at least two LLMs.

Evaluation measures business quality, not only text quality.

The rubric asks for traces and LLM comparison; the business needs proof that responses are safe, useful, reviewable, and worth their model cost.

Quality scorecard

■ Safety	No unsafe injury, supplement, fasting, or weight-cutting advice.
■ Relevance	Plan reflects age, sport, position, goal, equipment, and schedule.
■ Personalization	Uses profile context and matched development pattern.
■ Rejection	Gracefully refuses irrelevant or unsafe requests.
■ Review load	Flags injury cases and low-confidence outputs for humans.

Human-in-the-loop policy

Automatically route to review when:

1. injury or soreness appears
2. safety score is low
3. judge fails the response
4. request is adult-preview scope
5. model output conflicts with rules

The human does not replace the eval; the human validates the highest-risk decisions and generates feedback data.

The LLM choice is a unit economics decision.

The final model should be chosen by incremental value: does better safety and plan quality create enough business value to justify higher inference cost?

ROI formula

Incremental ROI =

$$\frac{(\text{extra business value} - \text{extra LLM cost})}{\text{extra LLM cost}}$$

Use final values from gold_llm_comparison_summary after running real endpoints.

Illustrative assignment example

Metric	Cheaper LLM	Premium LLM
Monthly model cost	\$200	\$400
Effectiveness	70%	84%
Successful plans on 10k/mo	7,000	8,400
Value per success	\$3	\$3
Business value	\$21,000	\$25,200

Premium creates +\$4,200 value at +\$200 cost -> 20x incremental ROI.

One academy can pay back the MVP; five academies make it a repeatable business.

The original proposal frames a practical academy subscription path that does not require full consumer scale to justify a pilot.

Academy subscription bridge

$$\begin{array}{ccccccc} \mathbf{250} & & \mathbf{\$12} & & \mathbf{\$3k} & & \mathbf{\$36k} \\ \text{athletes per academy} & \times & \text{per athlete per month} & = & \text{monthly revenue} & = & \text{annual revenue per academy} \end{array}$$

Proposal payback logic

\$20k MVP build + about \$800/month operating cost can be recovered in less than one year by a single 250-athlete academy.

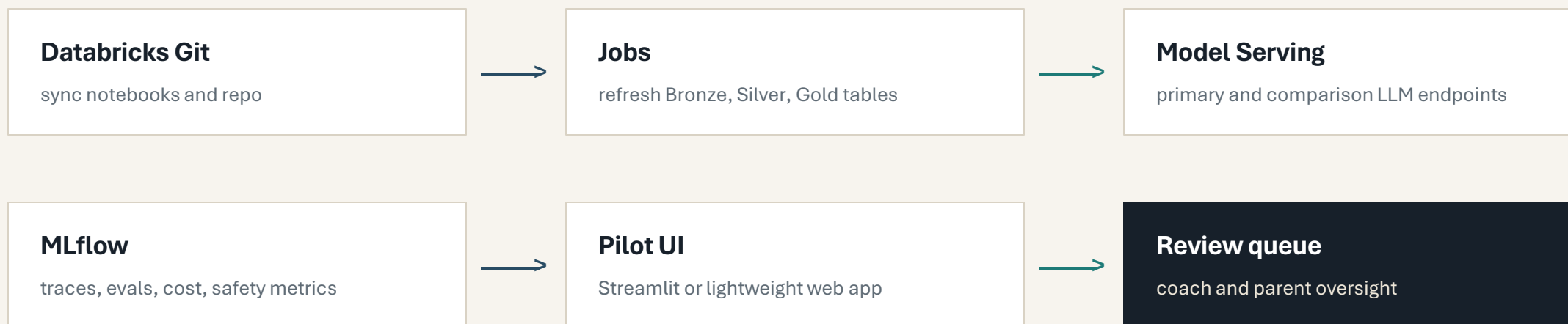
Expansion math

Five academies at the same price point imply roughly \$180k annual recurring revenue before operating and support costs.

Business value is not only revenue: the pilot also creates plan-completion data, soreness flags, coach ratings, and human edits that become the proprietary feedback loop.

Deployment should begin as a supervised academy pilot.

The recommended production path keeps Databricks as the quality backbone while the user experience starts simple.



Pilot cadence: 8 weeks with one academy. Gate expansion on safety pass rate, coach acceptance, plan completion, time saved, and retention signal.

The POC works, but the investable version needs intake and trace maturity.

This is the opinionated commentary the rubric asks for: what exceeded expectations, what did not, and what we learned.

Exceeded

Databricks pipeline creates agent-ready Gold tables across profiles, exercises, benchmarks, nutrition, safety rules, progress metrics, and feedback schema.

Needs work

The product should move beyond athlete ID lookup into free-form profile intake, validation, and similarity matching.

Risk

The current public-data POC has limited defensibility until academy pilots capture completion, soreness, rating, and human-edit feedback.

What we learned about agents

The hard part is not generating fluent plans. It is making the agent observable, evaluable, safe for minors, and economically justified model-by-model.

Pilot the safer advisor, then earn the data moat.

The final recommendation is to launch a supervised academy pilot only after real endpoint traces, profile intake, and human-review routing are visible in Databricks.

1 Finish the agent proof

Add profile intake, matching, and visible MLflow spans.

2 Run the model decision

Compare two real LLM endpoints on quality, latency, cost, and ROI.

3 Launch one academy pilot

Measure coach acceptance, completion, safety, and retained athletes.

Go

with supervised pilot

Submit the GitHub repo, video walkthrough, and Databricks notebook evidence as the technical artifact set.